

Greenlight greening the electricity sector

India's **path** to **decarbonisation** **lies** through its electricity sector. The country has been increasing its RE capacity - from 38.96 GW in 2015 to 203.18 GW in 2024. However, it must not only **accelerate** its **pace** to **meet** the **ambitious** target of 450 GW of RE by 2030 but also **roll out** a modern and smart transmission system. Equally important is **addressing** structural, regulatory and tariff-related challenges **associated** with integrating clean and **intermittent** energy sources. An **integrated** approach is crucial to ensuring a smooth and **equitable green transition**.

Beyond budgetary **interventions** for generation and transmission, GoI must initiate regulatory changes, revise market designs and reform tariff systems to **power** the transition and transformation of the electricity sector. Policy interventions should focus on **equipping** states to adopt integrated resource planning and **robust resource-adequacy** frameworks. These measures will guide investments in flexible, clean technologies, and modern transmission **infrastructure**.

Additionally, **changes** in market design **are** necessary to correct the **existing bias** towards coal. This requires innovative **mechanisms** to restructure **legacy** PPAs to enable an economic electricity dispatch system - one that respects existing contracts while **incentivising** new, cleaner providers. **Addressing** the economic **viability** of discoms **is** another critical challenge. Despite **numerous** rescue packages, **discoms** remain financially **unviable**. A fresh, **out-of-the-box** approach is required, integrating distributed RE, battery storage, smart meters and dynamic tariff designs to enable **sustainable** new business models. The budget must also **lay the groundwork** for a robust retail electricity market that values clean, **efficient** and cost-effective systems. Without structural and regulatory interventions, **greening** the electricity sector **will** remain a **formidable** challenge, **jeopardising** India's decarbonisation goals. **[Practice Exercise]**

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

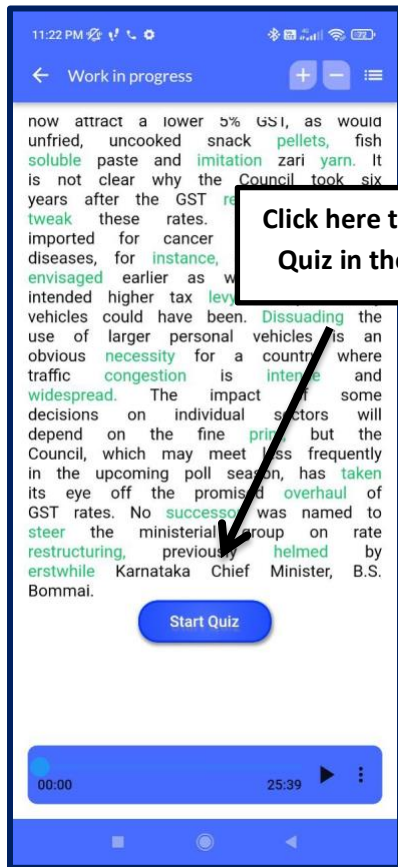
1. **Greenlight** (noun) – approval, clearance, go-ahead, sanction, consent अनुमति
2. **Decarbonisation** (noun) – reduction of carbon emissions, lowering carbon footprint, transition to low-carbon, cutting CO₂ output, removing carbon from processes कार्बन उत्सर्जन में कमी
3. **Lie** (verb) – be found, be situated, reside, exist, rest स्थित होना
4. **Accelerate** (verb) – hasten, speed up, quicken, expedite, increase pace तेज करना
5. **Pace** (noun) – speed, rate, tempo, velocity, momentum गति
6. **Meet** (verb) – fulfil, achieve, satisfy, comply with, match पूरा करना
7. **Ambitious** (adjective) – aspirational, challenging, daring, lofty, keen महत्वाकांक्षी
8. **Roll out** (phrasal verb) – launch, introduce, implement, deploy, bring out आरंभ करना
9. **Address** (verb) – tackle, deal with, attend to, confront, manage समाधान करना
10. **Associated** (with) (adjective) – connected with, linked to, related to, tied to, involved with से संबंधित
11. **Intermittent** (adjective) – sporadic, irregular, periodic, discontinuous, recurrent रुक-रुक कर आने वाला
12. **Integrated** (adjective) – combined, unified, cohesive, consolidated, joined एकीकृत
13. **Equitable** (adjective) – fair, just, unbiased, impartial, even-handed न्यायसंगत
14. **Green transition** (noun) – shift to sustainable development, movement toward clean energy, ecological transformation, environmental transition, shift to low-carbon economy हरित परिवर्तन
15. **Intervention** (noun) – involvement, intercession, mediation, action, interference हस्तक्षेप
16. **Power** (verb) – energise, drive, fuel, propel, empower शक्ति देना
17. **Equip** (verb) – furnish, provide, supply, arm, prepare सुसज्जित करना
18. **Robust** (adjective) – strong, resilient, sturdy, vigorous, solid मज़बूत

19. **Resource-adequacy** (adjective) – Pertaining to systems or frameworks ensuring sufficient resources (e.g., power capacity) to reliably meet demand. संसाधन-पर्याप्तता संबंधी
20. **Infrastructure** (noun) – framework, system, support facilities, physical structures, basic equipment बुनियादी ढांचा
21. **Existing** (adjective) – current, present, in place, prevailing, operative मौजूदा
22. **Bias** (noun) – partiality, prejudice, inclination, predisposition, favouritism पक्षपात
23. **Mechanism** (noun) – system, process, method, procedure, apparatus तंत्र / प्रणाली
24. **Legacy** (noun) – inheritance, heritage, tradition, consequence, aftermath विरासत
25. **Incentivise** (verb) – encourage, motivate, stimulate, spur, promote प्रोत्साहित करना
26. **Viability** (noun) – feasibility, practicability, sustainability, workability, profitability व्यवहार्यता
27. **Numerous** (adjective) – many, countless, plentiful, myriad, a multitude of अनेक
28. **Discom** (noun) – Short for “distribution company,” responsible for the supply and management of electricity in a region. बिजली वितरण कंपनी
29. **Unviable** (adjective) – not feasible, impractical, nonviable, infeasible, unsustainable अव्यवहार्य
30. **Out-of-the-box** (phrase) – innovative, creative, unconventional, novel, imaginative नए और रचनात्मक दृष्टिकोण से
31. **Sustainable** (adjective) – eco-friendly, viable, renewable, enduring, maintainable टिकाऊ
32. **Lay the groundwork** (phrase) – prepare the base, set the foundation, make preparations, pave the way, establish the basis आधार तैयार करना
33. **Efficient** (adjective) – effective, productive, capable, competent, proficient कुशल
34. **Formidable** (adjective) – challenging, difficult, daunting, intimidating, tough कठिन / दुष्कर
35. **Jeopardise** (verb) – endanger, put at risk, imperil, threaten, compromise खतरे में डालना

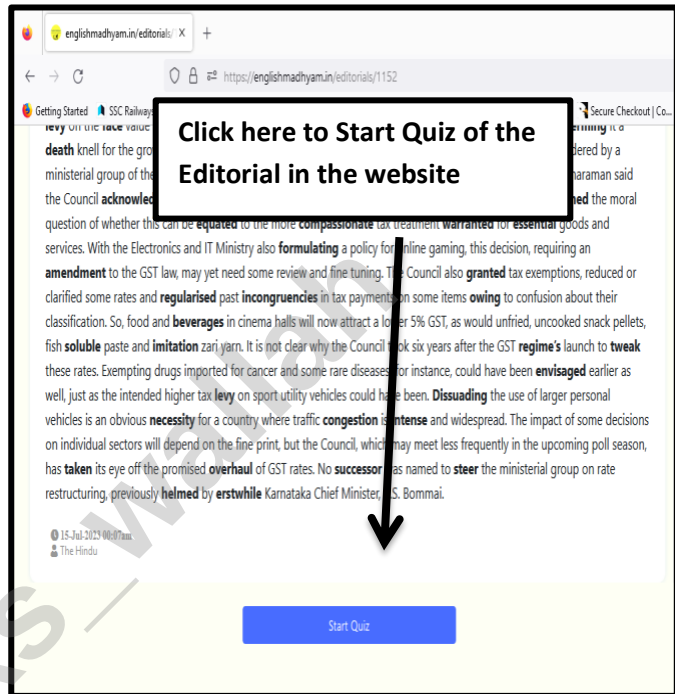
Summary of the Editorial

1. **Decarbonisation Focus:** India's path to decarbonisation is closely tied to reforming its electricity sector, particularly through renewable energy (RE).
2. **Increasing RE Capacity:** India's RE capacity has grown from 38.96 GW in 2015 to 203.18 GW in 2024, with a target of 450 GW by 2030.
3. **Modern Transmission Systems:** Achieving the 2030 target requires accelerating RE adoption and implementing a modern, smart transmission system.
4. **Addressing Challenges:** Structural, regulatory, and tariff-related challenges must be addressed to integrate clean, intermittent energy sources.
5. **Integrated Approach:** A comprehensive strategy is needed for a smooth and equitable green transition in the electricity sector.
6. **Regulatory Reforms:** Beyond budgetary allocations, the government must implement regulatory changes and revise market designs to support the transition.
7. **State-Level Frameworks:** States should adopt integrated resource planning and resource-adequacy frameworks to guide investments in clean technologies and infrastructure.
8. **Market Design Overhaul:** Current market designs favor coal; innovative mechanisms are needed to restructure legacy Power Purchase Agreements (PPAs) and incentivize cleaner energy providers.
9. **Economic Dispatch System:** India should transition to an economic electricity dispatch system that balances respect for existing contracts with incentives for clean energy.
10. **Discom Viability:** Financially troubled discoms remain a critical challenge, despite rescue packages.
11. **Innovative Solutions for Discoms:** Integrating distributed RE, battery storage, smart meters, and dynamic tariffs is essential to create sustainable business models for discoms.
12. **Retail Market Development:** A robust retail electricity market must be established, emphasizing clean, efficient, and cost-effective systems.
13. **Budgetary Interventions:** Policy measures should prioritize funding generation, transmission, and state-level adoption of innovative frameworks.
14. **Dynamic Tariff Designs:** New tariff systems should encourage flexibility and support clean energy adoption while addressing the financial challenges of stakeholders.
15. **Urgency of Reforms:** Without structural and regulatory interventions, India's green electricity transition will face significant obstacles, risking its decarbonisation goals.

Practice Exercise: Banking Pattern Based



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